

Unraveling the Impact of Data Breaches: Evidence From the US Healthcare Sector

Juhee Kwon¹  and M Eric Johnson²

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Abstract

As data breaches become an increasingly common risk, they are now a matter of “when” rather than “if.” While there has been active research on breaches, much of the work has focused on the antecedents and prevention of breaches, leaving the quantitative ramifications and variations in breach impact relatively unexplored. To address this gap, our study quantifies the impact of breaches on operational performance and identifies how operational, technological, and market factors moderate the impact. Using a quasi-experimental design with a difference-in-differences technique and propensity score matching, we analyzed a matched sample of 1,766 US hospitals, consisting of 883 breached hospitals and their non-breached peers from 2010 to 2017. We find that breached hospitals experience a 2.1% reduction in hospital admissions and a 0.28% decrease in market share. Moreover, network affiliation, decentralized governance, and cloud-based information technology services negatively moderate breach impacts, while IT security systems for detection, identity governance, and recovery provide mitigating effects. Additionally, we find that breaches in one hospital spillover to affect non-breached hospitals in the same local market. Our findings contribute to the operations management and security literature and provide managerial insights for enhancing breach resilience. Understanding these moderating factors can help hospital managers and policymakers formulate tailored mitigation strategies.

Keywords

Breach Resilience, Empirical Research, Healthcare Management, Information Security, Risk Management

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1 Introduction

The US healthcare sector has undergone rapid digitization over the past decade, fueled by governmental incentives promoting electronic health record (EHR) adoption. This digital transformation has reshaped practices, roles, interdependencies, and communication mechanisms among a diverse array of stakeholders, including hospitals, insurance companies, government agencies, and patients (McCullough et al., 2016). While digitization has enhanced care quality and efficiency, it has also expanded the spectrum of vulnerabilities due to the increased interconnectivity. According to a recent IBM report, a staggering 95% of surveyed organizations experienced more than one data breach. These breaches are not only becoming more frequent but also more impactful, leading to extended downtime of core business processes and substantial productivity losses (Soni et al., 2023). Particularly in healthcare, lucrative patient data attract both internal and external predators, making it a matter of time before a healthcare provider is victimized. As a result, information security strategies have been shifting beyond mere prevention,

increasingly emphasizing breach resilience to effectively mitigate impact and restore disrupted operations (Cisco, 2020).

There is a significant body of multidisciplinary research on information security. From the ex-ante perspective, security researchers have examined the effectiveness of security investments in preventing breaches (Herath and Herath, 2008) and the impact of security policies on deterring breaches (Burns et al., 2022). Operations management (OM) researchers have developed theoretical and empirical models to investigate

¹ College of Business, City University of Hong Kong, Kowloon Tong, Hong Kong

² Owen Graduate School of Management, Vanderbilt University, Nashville, TN, USA

Corresponding author:

Juhee Kwon, College of Business, City University of Hong Kong, Kowloon Tong, Hong Kong.

Email: juhee.kwon@cityu.edu.hk

diverse security strategies. Theoretically, they have developed analytical models to optimize security sourcing strategies (Cezar et al., 2017), design interorganizational contracts with third-party suppliers to improve security management (Jayaraman et al., 2013), and address the decision-making problem of cybersecurity risk planning, including resource allocation strategies for both government and business entities (Massimino et al., 2018). Empirically, they have investigated the effectiveness of reputational sanctions on reducing security negligence (Tang and Whinston, 2020) and the timing of vulnerability disclosures to prevent potential exploits that could harm installed systems and users (Sen et al., 2020). From the ex-post perspective, economic researchers have estimated stock price changes after breach announcements (Campbell et al., 2003). Information systems researchers have studied the impact of user compensation strategies as a data breach recovery action (Hoehle et al., 2022). Given the increasing importance of consumer data, marketing researchers have explored changes in consumer perceptions in the aftermath of a breach (Brough et al., 2022).

Despite this growing body of research, little is known about the quantitative ramifications of data breaches on organizational performance and how operational, technological, and market factors moderate breach impact. Noting this research gap, Massimino et al. (2018) called for an explicit examination of the relationship between digital confidentiality and diverse performance dimensions (e.g., productivity). Likewise, Kumar and Mallipeddi (2022) emphasized the importance of understanding the impact of cybersecurity on healthcare operations due to the sweeping digitization of healthcare information and processes. Our study aims to address this knowledge gap by quantifying the adverse impact of breaches on hospital performance and investigating how variations in operating structures shape these consequences. We draw on the OM and information security literature to formulate our research questions: (R1) how do data breaches impact hospital performance? and (R2) how do operational, technological, and market factors moderate the impact of breaches? By answering these questions, this study extends the extant literature by unpacking the pathways to achieving breach resilience. Our findings also offer valuable insights to hospital managers and policymakers working to develop incentives that nudge hospitals to tailor mitigation strategies to their operational and market environments.

To address the research questions, we analyzed 1,371 data breaches that occurred among 926 out of 4,696 US hospitals from 2010 to 2017. Given the diverse organizational characteristics influencing hospital performance (such as care quality, operational capabilities, and information system usage), our research context is susceptible to selection bias from confounding factors. To mitigate this potential bias, we employed a difference-in-differences (DID) technique combined with propensity score matching (PSM). Using this approach, we find that breached hospitals experience a 2.1% decrease¹ in admissions and a 0.28% loss in market share. To put the impact

in an economic perspective, a 2.1% decline translates to an average loss of about 60 inpatient visits per hospital, representing a \$1.6m loss in annual inpatient revenue.² We also find that organizational characteristics, such as network affiliation, decentralized governance structure, and cloud-based IT services, have a negative moderating impact. However, the adoption of IT security systems for detection, identity governance, and recovery mitigates breach impact. Furthermore, we find that hospitals can endure a breach without significant loss in highly concentrated markets. Notably, hospitals can dull the negative impact of a breach with restrictive health insurance plans that prevent patients from choosing hospitals based on their preferences.

Our study offers three important contributions to the existing literature as well as practical insights. First, it focuses on ex-post resilience strategies, distinguishing it from prior studies that emphasized ex-ante preventive strategies. Prior studies identified effective countermeasures to prevent security failures (Gal-Or and Ghose, 2005) and explored the likelihood of breach occurrence based on firm attributes such as age, size (Angst et al., 2017), operational maturity (Kwon and Johnson, 2013), and organizational (and governmental) policies (Tang and Whinston, 2020). While these prior studies enhanced knowledge about preventing and proactively addressing security risks, there has been a dearth of research on organizational breach resilience. This study quantifies the ramifications of breaches on hospital performance from both operational and market perspectives. By unraveling the consequences of increasingly widespread breaches and their moderating effects on hospital performance, our findings contribute to the OM and security literature. Second, this study extends the line of reasoning beyond security-specific factors to institutional and operational approaches. Specifically, we examine the moderating effects of a complex set of hospital operating factors, such as network affiliation (Kumar et al., 2022) and governance structure. We find that breaches in a hospital with network affiliation cause a greater reduction in performance compared to a single hospital system. This result makes sense because breaches in healthcare networks could expose patient data shared by member hospitals (as part of an integrated delivery system), requiring more complicated remediation processes. Additionally, decentralized governance, which often leads to operational complexity in coordinating autonomous hospitals, has a negative moderating impact. Third, this study sheds light on critical nuances for IT strategies by highlighting the moderating impacts of IT sourcing and security system structures. The findings reveal that hospitals embracing cloud-based IT services experience more pronounced adverse effects from breaches. This is likely attributed to the intricacies of remediation involving interconnected process recovery and coordination with cloud service providers (Yang et al., 2021). We also highlight the efficacy of adopting a well-balanced mix of IT security systems in mitigating breach impact, as substantiated by significant findings across various IT security systems.

2 Theoretical Background and Hypotheses

In this section, we explore mechanisms linking data breaches to hospital performance and identify how breach impacts vary with operational and technological factors. In services as complex as healthcare, it is challenging to establish a single unifying theory that outlines the mechanisms connecting breaches and outcomes because of the complexity arising from both external factors (i.e., regulations and heterogeneous regional markets) and internal factors (i.e., operating, governance, and IT service structures) (Acquisti et al., 2016). We thus develop mechanisms by drawing from multiple theoretical backgrounds, including the OM and security literature, while considering the regulatory and economic context of the US healthcare sector.

2.1 Operational and Legal Challenges

The US healthcare sector is one of the most highly regulated industries. The Health Information Technology for Economic and Clinical Health (HITECH) Act of 2009 facilitated the transition from paper records to EHRs, but also escalated the enforcement of the Health Insurance Portability and Accountability Act (HIPAA) with strict breach notification rules. Violations of HIPAA rules can result in financial penalties; thus, breaches expose hospitals to legal challenges as well as operational disruptions.

Data breaches often bring about significant operational challenges (Cisco, 2020). For instance, malware or ransomware attacks disrupt a hospital's EHRs, electronic devices, or servers, rendering patient medical histories inaccessible and electronic patient monitoring inoperable. These disrupted processes can result in significant delays in healthcare services, eroding care delivery (Anderson and Agarwal, 2011). Security researchers have observed that complicated risk management processes lead to prolonged operational disruption, potentially resulting in reputational damage (Kim et al., 2008). OM researchers have demonstrated how operational failures impact productivity and financial outcomes across diverse organizational structures and control capabilities. Fianu et al. (2022) argued that operational failures cause imbalances between supply and demand, leading to a proliferation of risks that impact productivity and spillover into other areas. Masimino et al. (2018) suggested that as digital assets continue to expand in scale and scope, the increasing risk to digital confidentiality poses a serious threat to operations and supply chain management.

Moreover, breached hospitals must demonstrate commitment to taking remedial steps and maintaining ongoing HIPAA compliance (Hu et al., 2013). HIPAA rules mandate that breached hospitals conduct immediate breach notifications and take corrective actions such as performing forensic and investigative activities, assessments and audits, and crisis management. Based on the extent of a breach, hospitals must further ensure the security of the data environment, including their business associates (Straub and Welke, 1998). If a

breach affects information shared with a hospital's business associates (or is caused by them), the hospital should confirm that appropriate security agreements are in place, update those agreements if necessary. In cases where a breach is caused by careless employees, hospitals may need to consider measures such as terminating negligent employees or providing them with remedial training on healthcare data handling and security practices (Willison and Warkentin, 2013).

Managing these burdens, essential for restoring operational disruption and meeting regulatory requirements, takes considerable time and staff effort. The remediation phase may extend over months or years as attention shifts toward repairing damage to the business and preventing similar occurrences in the future. Consequently, breach remediation can lead to operational overhead and a significant loss in performance (Schmidt and Raman, 2022). We thus hypothesize:

H1: A data breach will have a negative impact on hospital performance.

2.2 Operating Structure

As digitization advances and intensifies market competition, US hospitals have undergone significant operational and technological changes. To enhance competitiveness, many hospitals have joined healthcare networks, facilitating expanded care delivery (Cherkesly et al., 2019). The governance structure within these expanded organizational frameworks has been transformed. Technologically, hospitals have adopted cloud-based IT services to access cutting-edge capabilities and have implemented IT security systems for data protection (Guha and Kumar, 2018). This operational and technological expansion has enlarged the cyber-attack surface and increased partnership complexity, resulting in a growing impact of breaches and an escalating need for remediation. In the next section, we explore how these changes affect breach impact.

2.2.1 Healthcare Service Structure. Healthcare networks are a dominant operational structure in the USA, comprising a corporate center and two or more affiliated providers managed under consolidated structures to integrate healthcare service delivery. Approximately 67% of US hospitals are affiliated with healthcare networks (AHA, 2023). In these networks, various types of organizations collaborate to provide patient care across the care continuum, including primary, secondary, and tertiary care facilities (Kumar et al., 2022). To achieve integrated service delivery, healthcare services and supporting IT systems must be interconnected, leading to increased operational and IT complexity (Mocker et al., 2014).

While healthcare networks offer numerous benefits, like seamless care delivery, cost-effectiveness, and an enhanced ability to adapt to a changing healthcare environment (Lu and Lu, 2018), they pose significant security challenges. Information sharing across member hospitals can amplify the impact of breaches, creating pathways for breaches to affect multiple hospitals simultaneously. The extended scope of healthcare

services and information sharing renders more access points to patient data, thereby enlarging the attack surface. It is more challenging to coordinate the security controls for a larger scope of access points (Fianu et al., 2022). For instance, in a malware outbreak, a victim hospital would be cut off by other healthcare providers, who could refuse to reconnect the hospital until its network is declared completely clean and malware-free by an independent forensic consultant. Meanwhile, the breached hospital must independently provide health services and bear the associated costs needed to operate. This means that breaches in a healthcare network not only directly erode the performance of a breached hospital, but also indirectly hurt performance by requiring the hospital to conduct extensive breach detection, recovery, and remediation before rejoining the network. The larger attack surface and interconnectivity of a healthcare network present additional challenges in operational and legal remediation.

Despite these challenges, some scholars and practitioners argue that healthcare networks enjoy significant security advantages over individual hospitals (Radanliev and De Roure, 2022; Soni et al., 2023). Large healthcare networks are more able to support teams of technological and human experts capable of swiftly responding to and mitigating the impact of a breach. The resulting mobilization of a network's resources can effectively address damages across the entire network, leading to a more robust and coordinated response. Furthermore, remediation efforts within a healthcare network can leverage economies of scale. A network's access to advanced security tools, technologies, and services is likely well beyond the financial reach of individual hospitals. This capability can enhance the overall security posture of the network, improving its effectiveness in remediating breaches. Additionally, healthcare networks often enforce standardized security protocols across all network members. This standardization facilitates a faster and more uniform response to a breach, guided by established protocols familiar to all network participants. With these opposing perspectives, we present two competing hypotheses below:

H2a: A hospital's network affiliation will exacerbate the negative impact of a data breach on hospital performance.

H2b: A hospital's network affiliation will alleviate the negative impact of a data breach on hospital performance.

2.2.2 Decentralized Governance. A hospital's network affiliation positions it within the governance structure (centralized or decentralized) of its network, thus exerting influence over the operations and decision-making processes of the hospital. While decentralized governance provides autonomy to member hospitals (Bazzoli et al., 1999), centralized governance grants a corporate center the legal or fiduciary authority to operate and control its member hospitals. The governance structure shapes operational complexity in information sharing and service integration (Ding and Kaminsky, 2020; Holmstrom and Milgrom, 1991), impacting the extent

to which a breach disrupts operations and how rapidly and effectively breach remediation activities can be conducted.

Prior research suggested both advantages and disadvantages of centralized and decentralized governance (Tiwana and Kim, 2015). Guo et al. (2019) demonstrated that decentralized strategies can boost productivity through healthy competition among member organizations. Xue et al. (2011) proposed that decentralized approaches empower member organizations to make informed decisions tailored to local needs, changing conditions, and new opportunities. Local empowerment provides flexibility for nuanced and rapid responses, particularly in dealing with localized breaches or department-specific issues (Nault, 1998; Xue et al., 2011). Such flexibility and discretion also allow member hospitals to leverage local expertise and customize solutions for the unique challenges they face.

On the contrary, Holmstrom and Milgrom (1991) argued that decentralized governance poses challenges in managing diverse risks when the goals of member organizations do not perfectly align, whereas centralized governance can effectively reconcile differences and manage common risks. The centralized approach introduces coordination and uniformity challenges, particularly in terms of recovering from breaches and implementing remedies. Under decentralized governance, member hospitals may prioritize mitigation and compliance differently during a breach, leading to time-consuming negotiations for forensic and investigative activities in interconnected operations (Flores et al., 2014). Moreover, variation in system protocols among members can increase technical complexity, potentially prolonging breach remediation and introducing security risks (Majchrzak et al., 2015). Recovering from a breach and remediating its damage requires establishing or rebuilding overarching organizational norms regarding IT structure, security controls, compliance, and security mindsets. Thus, centralized governance can streamline decision-making and problem-solving through standardized methods and activities, fostering efficient and coordinated responses to breaches with a unified approach to security. These previous studies collectively suggest that decentralized governance enhances strategic agility at the local level but may introduce coordination challenges and increased technical complexity, whereas centralized governance provides efficiency and coordination but may lack local adaptability (Tiwana and Kim, 2015). This research background sets the stage for the exploration of two competing hypotheses below:

H3a: A decentralized governance structure of a hospital's network will exacerbate the negative impact of a data breach on hospital performance.

H3b: A decentralized governance structure of a hospital's network will alleviate the negative impact of a data breach on hospital performance.

2.2.3 Cloud-Based IT Services. With the rapid escalation of technologies and market competition, hospitals have grappled with how to maintain cost-effectiveness and enhance advanced care services. Cloud-based IT services have emerged as a

solution, offering cutting-edge technologies, flexibility, and scalability (Yang et al., 2021). Many hospitals have rushed to adopt cloud-based IT services for technical agility and cost savings (Zhang et al., 2020). However, this shift comes with increased security risks due to the growing complexities of IT service structures and the challenges of securely processing sensitive data in the cloud (Barthelemy, 2001).

Despite such security concerns, hospitals have been increasingly adopting cloud-based IT services for a broader range of functions and storage needs. According to an industry survey, 84% of hospitals reported the use of cloud-based IT services for multiple functions, such as analytics, healthcare services, finance, and human resources. Nevertheless, the respondents also acknowledged that the use of remote-hosted resources and processes diminishes their control over confidential information, hindering swift remediation (August et al., 2014). The responsibility for maintaining security protocols, complying with privacy and security regulations, and monitoring for breaches or vulnerabilities is often unclear when using cloud-based IT services. This shared responsibility complicates determining actions to mitigate breaches and, in the case of penalties, deciding who pays, how much, and when (Yang et al., 2021).

To identify the root causes of a breach and prevent similar future incidents, client organizations must assess the security measures taken by cloud service providers. This involves auditing cloud-stored data and scrutinizing third-party agreements for compliance with HIPAA rules. The proliferation of cloud-based IT services and associated security concerns has led to increased regulatory burdens. The US Health and Human Services (HHS) has issued guidance categorizing cloud service providers as business associates subject to HIPAA rules (HHS, 2022). Compliance with regulatory requirements and ensuring service security are the topmost considerations when utilizing cloud-based IT services, which necessitates close coordination among cloud service providers and between them and client organizations. As transaction complexity escalates, so do the coordination costs of care processes. The growing risks and legal complexity associated with cloud-based IT services can slow breach remediation, potentially degrading hospital performance. Therefore, we hypothesize:

H4: A hospital's reliance on cloud-based IT services will exacerbate the negative impact of a data breach on hospital performance.

2.2.4 IT Security Systems. Considerable prior studies have explored the impact of IT security systems on breach occurrences. Building on the theory that more resources lead to better performance, those studies demonstrated that the adoption of IT security systems can enhance data protection (Angst et al., 2017; Kwon and Johnson, 2014), and reduce security violations (Cram et al., 2019). While this research stream considered IT security systems primarily as defenders against

breaches, these systems can also facilitate breach mitigation, reducing impact.

In a post-breach period, IT security systems operating across disparate systems can detect intrusions, track such intrusions, differentiate between authorized and unauthorized activities, and recover compromised data and processes (Bensoussan et al., 2020). Even if a breach disrupts operations or unauthorized individuals gain access to the data, IT security systems can expedite the restoration of the disrupted operations by reducing post-breach downtime through planned backup processes and accelerating the identification of any process, file, or application altered during the breach. In healthcare, Tucker (2004) argued that operational failures in a hospital do not always result in delayed healthcare services if backup and recovery functions work effectively, and such operational failures with well-established recovery processes can provide employees with valuable learning opportunities. Quickly finding and managing compromised IT surfaces can help organizations enhance their security posture by locating previously unknown vulnerabilities (Tucker, 2004). Moreover, managing regulatory remediation requirements becomes more attainable when organizations promptly identify the root causes of breaches during the recovery process. Thus, hospitals equipped with robust IT security systems can expedite recovery processes, efficiently allocating fewer resources per breach remediation. This strategic approach not only ensures swift resolution but also enables the maintenance of uninterrupted care services. This forms the foundation for our hypothesis:

H5: A hospital's IT security systems will alleviate the negative impact of a data breach on hospital performance.

3 Research Methods

3.1 Data Collection

We compiled panel data on US hospital information and data breaches from multiple sources. Hospital information was obtained from three sources: the Healthcare Information and Management Systems Society (HIMSS) Analytics™ Database, the American Hospital Association (AHA), and the Centers for Medicare and Medicaid Services (CMS). The HIMSS database provides comprehensive data on hospitals' IT systems, for example, clinical decision support systems (CDSS), computerized provider order entry (CPOE) systems, EHRs, security software, and cardiology/laboratory IT systems. The AHA database includes admissions, governance structure, health insurance plans, network affiliation, and basic demographics (e.g., bed sizes and geographic market units). As a federal government agency, the CMS provides hospital-level quality data and incentive payment information related to EHR meaningful use attestation under the HITECH Act. These datasets were merged using hospital Medicare identification numbers. Focusing on acute-care hospitals that provide short-term medical and/or surgical care, our total sample includes 4,696 hospitals from 2010 to 2017.

Next, healthcare breaches were collected from two sources: HHS³ and the Privacy Rights Clearinghouse (PRC). HHS has maintained a list of breaches under HIPAA rules on its so-called “Wall of Shame” website since 2009. PRC, a California non-profit organization, has published breach information across all industries since 2005. Both sources include brief descriptions of each case, noting the number of breached records, breach date, type, and healthcare provider details. From 2010 to 2017, a total of 1,371 breaches occurred, comprising 609 internal, 730 external, and 32 breaches with unknown sources, impacting 926 out of 4,696 hospitals.

3.2 Operationalization of Variables

In this study, breaches are categorized into two types: *external* and *internal*. While external breaches are carried out by cybercriminals aiming for financial gain through identity theft, device lockouts, or data blockages, internal breaches are more likely the result of internal mistakes and organizational failures. This distinction is important because the impact of a breach can vary depending on its type. Based on breach descriptions from HHS and PRC, a breach with terms like *theft*, *stolen*, *hacker*, *ransomware*, *malware*, and *virus* is categorized as *External*, while one with terms like *loss*, *improper disposal*, *accidental exposure*, and *errors* is categorized as *Internal*.

Hospital performance is measured by two major factors: hospital admissions (*Admissions*) and market share (*MKshare*). *Admissions* represents the log-transformed inpatient care volume. Managing admitted patients necessitates coordinated efforts in integrated healthcare and IT services, including scheduling, capacity planning, resource allocation, and intensive care. Thus, hospital admissions serve as a meaningful measure for evaluating hospital performance. *MKshare* denotes the percentage of the inpatient care market attributed to a hospital. It is calculated by dividing a hospital's admissions by the total admissions of all hospitals in its healthcare referral region (HRR).⁴ Evaluating the change in a breached hospital's market share illuminates changes in its competitive position, reflecting shifts in patient preferences and choices, reputational damage, and operational disruptions relative to other hospitals in the HRR. To further explore hospital operating mechanisms, we employ two additional performance measures: average inpatient days (*AvgPatientDays*) and the logged number of emergency room (ER) visits (*ER visits*). To investigate the spillover effect of a breach in a local healthcare market, we incorporate the logged number of admissions for non-breached hospitals within a focal hospital's HRR (*AdmNoBreach*).

Our research models include multiple independent variables. *HealthNetwork* is a binary variable where one represents hospital affiliation with a healthcare network and zero denotes no affiliation. *Decentralized* measures the level of decentralized decision-making within healthcare networks on various issues such as healthcare services, resource allocation, and

physician and insurance arrangements. It ranges from most centralized (score of 1) to most decentralized (score of 5) and is based on the taxonomy developed by Bazzoli et al. (1999).⁵ *CloudITSvc* is the proportion of remote-hosted IT services to a hospital's total IT services. *ITSecurity* quantifies a hospital's IT security systems by counting the adopted security applications, categorized into four types: detection, prevention, identity governance, and recovery. Market competitiveness (*MKHHI*) is gauged using the Herfindahl–Hirschman index (HHI), which ranges from zero to one, moving from a large number of small hospitals to a single monopolistic hospital. Health insurance plans, affecting hospital choice, are classified into health maintenance organizations (HMOs), preferred provider organizations (PPOs), and fee-for-service (FFS) plans. The AHA database provides information on whether a hospital has contracts for each type of health insurance plan at different levels of the organization (e.g., hospital, health network, and joint venture) (AHA, 2017). The variable, *FFS*, depicts a hospital's contracts with FFS plans, reflecting insurance flexibility. Detailed information about the types of IT security systems and three health insurance plans can be found in Supporting Information EC1.

Following prior healthcare studies, our empirical models include diverse controls to account for potential confounding effects on hospital performance. First, as digital capabilities can affect both breach occurrence and hospital performance, we measure a hospital's digitization with two factors: EHR systems (*EHRs*) and IT applications (*ITapps*) (Kim and Kwon, 2019). The variable, *EHRs*, is measured by the number of applications, including CDSS, CPOE, and EMRs. *ITapps* is the number of IT applications related to healthcare and business functions such as cardiology, laboratory, transcription, finance, revenue cycle, and supply chain management. Since the meaningful utilization of EHRs beyond simple implementation can also affect security and hospital performance, we include the incentives that a hospital received after attesting to the EHR Meaningful Use (MU) program (*MU2*).

Next, in this research context we must control for care quality, which can significantly affect hospital performance as reflected in patient volume. We adopt standardized readmission rates, which are formally included in CMS's reimbursement decisions⁶ as a common measure of care quality (Andritsos and Tang, 2014). *Readmission* has a value zero if a hospital's readmission rate is worse than the US national rate, one if its rate is no different from the US national rate, and two if its rate is better than the US national rate. To control for scale, we incorporate the logged number of licensed beds (*Beds*) at a hospital, reflecting its size. The complexity of a hospital's healthcare service is controlled by including the number of healthcare services provided under agreements with external organizations (*HealthSvc*).⁷ To consider market power, we utilize the rank of a hospital's inpatient care volume (*HRRrank*) within its HRR. We further control for market characteristics related to information security by measuring the number of other hospitals that experienced breaches

Table 1. Variable definition and summary statistics.

Variable	Description	Mean	SD	Min	Max	Source
Data breaches						
TotalBreaches	Number of total breaches	0.034	0.190	0.00	2.00	HHS, PRC
External	Number of breaches caused by external attackers	0.017	0.131	0.00	2.00	
Internal	Number of accidental breaches caused by careless insiders	0.016	0.132	0.00	2.00	
Performance						
Admissions	Logged number of admissions	7.958	1.591	0.69	11.9	AHA
MKshare	Market share based on admissions	0.070	0.119	0.00 ^a	1.00	
AvgPatientDays	Average inpatient days per admission	5.501	11.88	0.00	352.8	
ERVisits	Logged number of emergency room visits	6.927	4.524	0.00	12.53	
AdmNoBreach	Logged number of admissions for non-breached hospitals within an HRR	10.65	2.173	0.00	13.4	
Independent variables (moderating factors)						
HealthNetwork	One if a hospital is a member of a healthcare network; zero otherwise	0.614	0.487	0.00	1.00	HIMSS
Decentralized	One if a hospital's system is most highly centralized and five if it is most highly decentralized ⁵	3.445	1.016	1.00	5.00	AHA
CloudITSvc	Ratio of cloud-based IT applications over total IT applications	0.008	0.015	0.00	0.35	HIMSS
ITSecurity	Number of adopted IT security applications	4.199	2.824	0.00	23.0	HIMSS
MKHHI	One if an HRR's HHI is greater than the average HHI, 0.177; zero otherwise	0.378	0.485	0.00	1.00	AHA
FFS	Number of fee-for-service plans at different levels of the organization (e.g., hospital, health network, and joint venture)	0.055	0.259	0.00	4.00	AHA
Control variables						
EHRs	Number of adopted EHRs (e.g., EMR, COPE, medical terminology, and data repository)	8.771	4.508	0.00	35.0	HIMSS
Itapps	Number of IT applications	57.19	17.83	1.00	139.0	HIMSS
MU2	Amount of incentives for a hospital's EHR meaningful use attestation	0.261	0.587	0.00	20.3	CMS
Readmission	Zero for a hospital with a higher readmission rate than the national average, one for a rate equal to the national average, and two for a rate better than the national average	0.494	0.510	0.00	2.00	CMS
Beds	Logged number of licensed beds	4.554	1.184	0.69	7.65	HIMSS
HealthSvc	Number of healthcare services provided under agreements with other organizations	3.309	3.449	0.00	65.0	CMS
HRRRank	Ranking based on the market share derived from admissions	13.29	13.16	1.00	84.0	AHA
HRRPeerBreaches	Number of other hospitals that experienced a breach within an HRR	0.926	2.096	0.00	16.0	HHS, PRC
NetPeerBreaches	Number of peer hospitals that experienced a breach within a network	0.985	7.926	0.00	107.0	

Notes: $n = 35,178$. Hospitals = 4,696.

^aThe minimum market share is 0.000006311418.

within a hospital's HRR (*HRRPeerBreaches*). The number of peer hospitals that experienced breaches within a hospital's network (*NetPeerBreaches*) is also employed to control for spillover. Table 1 provides the definitions and summary statistics of all variables in the total sample.

3.3 Identification Strategies

Given our research purpose of understanding how breaches impact hospital performance and the factors that moderate the impact, endogeneity issues must be considered (Heckman

et al., 1997). First, hospitals with low performance are likely to have limited resources, resulting in restricted security investments and, consequently, more breaches. This reverse causality suggests that breached hospitals may have underperformed compared to non-breached ones both before and after the breach. Second, hospital-specific latent factors, such as a well-managed workflow, skilled staff, and a well-established IT mindset, can influence both breach occurrence and hospital performance. For instance, hospitals with a well-established IT mindset might prioritize strong security and achieve higher performance through technology-enabled healthcare services

and security countermeasures. In this case, breach occurrence is correlated with hospital performance, but it does not directly affect performance.

To resolve these endogeneity issues, we employ PSM combined with a DID approach (Imbens and Wooldridge, 2009). First, PSM identifies changes in the post-treatment period between treatment and control groups with similar organizational covariates in the pre-treatment period. We create two groups: a “treatment” group of hospitals that experienced a breach and a “control” group of hospitals that did not. The control group serves as a counterfactual to assess what would have happened in the treatment group without a breach. Second, the DID configuration quantifies differences in pre- and post-treatment outcomes to account for hospital-specific variations (Wooldridge, 2010). By enabling each hospital to have its own baseline characteristics, we can determine relative changes in hospital performance within a hospital between the pre- and post-breach periods. If other important characteristics affect the differences in hospital performance, the changes in a breached hospital’s performance before and after a breach would be indistinguishable from those of its non-breached peer. After accounting for hospital characteristics, differences in hospital performance between treatment (breached) and control (non-breached) hospitals after a breach can be attributed to the breach. This approach considerably mitigates endogeneity issues and is more efficient than using instrumental variables that obviate the need for strong exclusion restrictions.

3.4 PSM–DID Model Specification

Our treatment group comprises breached hospitals, while the control group includes comparable non-breached peers in terms of hospital characteristics. We construct our PSM–DID models to account for both hospital and time effects (Wooldridge, 2010). Equation (1) addresses the direct breach impact on performance, while equation (2) explores moderating effects through an interaction term:

$$\text{Performance}_{it} = h_i + \alpha_1 \cdot \text{Treat}_{it} + \Gamma \cdot I_{it-1} + \Omega \cdot C_{it-1} + \Theta \cdot Z_{t-1} + y_t + \epsilon_{it} \quad (1)$$

$$\text{Performance}_{it} = h_i + \alpha_1 \cdot \text{Treat}_{it} + \alpha_2 \cdot (\text{TREAT}_{it} \times I_{it-1}) + \Gamma \cdot I_{it-1} + \Omega \cdot C_{it-1} + \Theta \cdot Z_{t-1} + y_t + \epsilon_{it} \quad (2)$$

where i denotes a hospital. Performance_{it} represents hospital i ’s performance at time t . In equation (1), the main parameter of interest is α_1 , capturing changes in the performance of breached hospitals relative to those of non-breached hospitals. TREAT_{it} is a treatment dummy variable, equal to one if hospital i has been breached at time t , and zero otherwise (Greene, 2011; Wooldridge, 2010). The specification includes hospital- and time-fixed effects. Hospital-fixed effects (h_i) account for time-invariant heterogeneity, and year dummies (y_t) handle time-fixed effects. I_{it-1} denotes the vector of the independent

variables with the coefficients (Γ). C_{it-1} represents the vector of a hospital’s control variables, Ω is the corresponding estimated coefficient, and Z_{t-1} is the vector of a healthcare market’s characteristics with coefficients (Θ). Standard errors are clustered at the hospital level. In equation (2), the parameter of the interaction term, α_2 depicts the heterogeneous impact of a breach.

4 Empirical Analyses and Results

We first assess the correlations between all explanatory variables. Supporting Information EC2 displays the correlation matrix with each variable’s tolerance (Tol) and variance inflation factor (VIF). While we find some elevated correlations between some variables (e.g., *Beds*, *Readmission*, *ITapps*, and *EHRs*), our multicollinearity diagnostics indicate that all Tols fall between 0.33 and 0.99, surpassing the common cutoff threshold of 0.1 (Hair et al., 2005). All VIFs range between 1.01 and 3.07, which is below the typical threshold of 10.0. We thus conclude that multicollinearity is not a significant concern in this study.

4.1 Constructing the Matched Sample

To construct comparable treatment and control groups, we utilize logistic regression to generate a hospital’s propensity score, predicting the likelihood of experiencing a breach based on organizational covariates. Detailed model specifications and results are provided in Supporting Information EC3. Using the estimated propensity scores, we employ a caliper matching technique to match a breached hospital with its non-breached peer (Rosenbaum, 2002). The absolute difference between the two propensity scores of a matched pair is defined as the “matching distance.” The matches are made within the 0.10 matching distance.⁸ Among 926 breached hospitals, 883 were successfully paired with non-breached counterparts, resulting in a matched sample of 1,766 hospitals.

To ensure matching quality, we conduct t -tests comparing variables between the total and matched samples in the pre-breach period, as shown in Supporting Information EC4. While all variables in the total sample display significant t -values, those in the matched sample do not. This indicates that the differences in covariates between the two groups in the matched sample are substantially smaller than in the total sample, suggesting a high degree of statistical equivalence in their characteristics. We further compare the differences in propensity scores to determine the extent of overlap in the characteristics of the treatment and control groups. In the total sample, a significant difference in the propensity scores for breach occurrence is observed between the two groups (3.85% for the control group and 6.02% for the treatment groups, with a t -value of -30.29). In the matched sample, the two groups are nearly identical (6.13% for the control group and 6.21% for the treatment groups, with a t -value of -0.31). Figure 1 depicts the density plots of the propensity scores for the two samples.

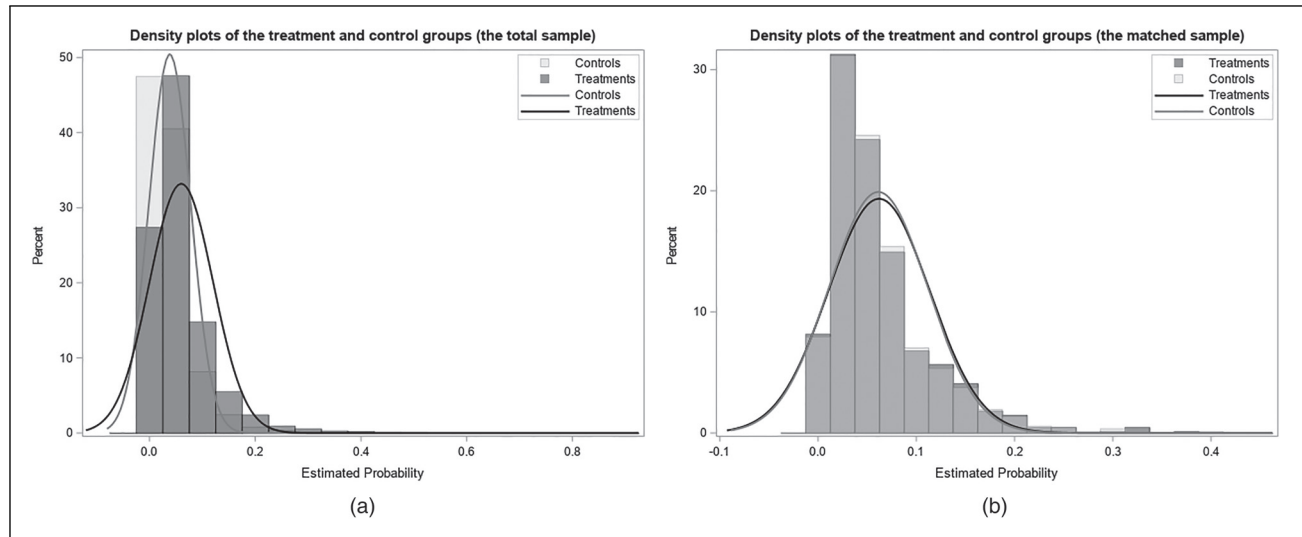


Figure 1. Density plots of propensity scores: propensity scores of (a) the total sample and (b) the matched sample.

Table 2. Impact of a hospital's breach on admissions.

Variable	(1)	(2)	(3)	(4)	(5)	(6)
Treat	-0.0213*** (0.008)	0.0126 (0.007)	0.0088 (0.007)	-0.0119 (0.009)	-0.0273** (0.012)**	0.0076 (0.008)
Treat × Network		-0.0335*** (0.011)				-0.0275** (0.012)
Treat × Decentral			-0.0070*** (0.002)			-0.0062** (0.003)
Treat × CloudITSvc				-1.1691** (0.537)		-0.8192*** (0.267)
Treat × ITSecurity					0.0034** (0.001)	0.0055*** (0.002)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Year-fixed effect	Yes	Yes	Yes	Yes	Yes	Yes
Hospital-fixed effect	Yes	Yes	Yes	Yes	Yes	Yes
R ² (within)	0.1314	0.1315	0.1322	0.1324	0.1312	0.1348
R ² (overall)	0.6144	0.6158	0.6137	0.6161	0.6145	0.6350

Notes: $N = 15,031$. Number of hospitals = 1,766. Standard errors in parentheses are clustered at the hospital level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

4.2 PSM-DID Results

Table 2 presents the effect of breaches and the moderating effects of diverse operating factors on the logged number of hospital admissions. In column (1) using equation (1), the treatment effect ($TREAT_{it}$) on admissions is significantly negative (-0.0213 at $p < 0.01$). Hospitals with a breach occurrence experience a 2.1% decrease in admissions, which translates to an estimated average loss of about 60 inpatients and a \$1.6m loss in annual inpatient revenue on average.² Thus, H1 is supported.

Columns (2) to (6) in Table 2, using equation (2), involve the interaction terms ($TREAT_{it} \times I_{it-1}$) to investigate heterogeneous breach impacts with diverse operating factors. Column (2) indicates that hospitals with network affiliation experience

a 3.3% decrease in admissions (-0.0335 at $p < 0.01$), translating to about 94 fewer admissions and a \$2.5m annual revenue loss. This supports H2a, but not H2b. Column (3) reveals a negative coefficient for the interaction term with decentralized governance (-0.0070 at $p < 0.01$). Hospitals with the most decentralized governance (score of 5), experience a 3.4% decrease in admissions, translating to 98 fewer admissions and a \$2.7m revenue loss, supporting H3a but not H3b. Columns (4) and (5) examine the moderating effects of a hospital's cloud-based IT services and IT security systems, respectively. The interaction term with the ratio of cloud-based IT services is negative (-1.1691 at $p < 0.05$), indicating that a 1% increase in the ratio translates to 33 fewer admissions and \$0.89m in annual inpatient revenue, supporting H4. On the other hand, IT security systems alleviate the negative effect with a 0.34%

Table 3. Impact of a hospital's breach on market share.

Variable	(1)	(2)	(3)	(4)	(5)	(6)
Treat	−0.0028** (0.001)	0.0010 (0.000)	0.0013 (0.000)	−0.0015 (0.001)	−0.0081*** (0.002)	0.0012 (0.001)
Treat × Network		−0.0033*** (0.001)				−0.0053** (0.002)
Treat × Decentral			−0.0008*** (0.000)			−0.0015** (0.000)
Treat × CloudITSvc				−0.1463** (0.077)		−0.1844** (0.091)
Treat × ITSecurity					0.0008*** (0.000)	0.0010** (0.000)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Year-fixed effect	Yes	Yes	Yes	Yes	Yes	Yes
Hospital-fixed effect	Yes	Yes	Yes	Yes	Yes	Yes
R ² (within)	0.0840	0.0879	0.0868	0.0856	0.0882	0.0860
R ² (overall)	0.3745	0.3623	0.3645	0.3836	0.3683	0.3711

Notes: N = 15,031. Number of hospitals = 1,766. Standard errors in parentheses are clustered at the hospital level. * $p < .1$, ** $p < .05$, *** $p < .01$.

increase (0.0034 at $p < 0.05$), translating to about 10 additional admissions and a \$0.27m increase in annual revenue. Column (6), which incorporates all interaction terms, yields consistent results.

We further analyze the impact of breaches on a hospital's market share to assess changes in patient choices compared to non-breached peers. As described in Table 3, the market share results align with those in admissions from Table 2. A breach reduces a hospital's market share by 0.28% (−0.0028 at $p < 0.05$). Hospitals with network affiliation experience a significant decrease in market share (−0.0033 at $p < 0.01$). The most decentralized governance and the ratio of cloud-based IT services also exhibit negative moderating effects (−0.0008 at $p < 0.01$ and −0.1463 at $p < 0.05$, respectively). However, adopting additional IT security systems mitigates the negative impact of breaches on market share by 0.08% (0.0008 at $p < 0.01$), consistent with the results in Table 2.

4.3 Robustness Tests

4.3.1 Leads and Lags Analysis. Using the leads and lags analysis (Autor, 2003), we examine whether there are differential pre-trends in the outcomes of interest between breached and non-breached hospitals. We create dummy variables indicating relative times to breach occurrence as follows: one dummy indicator for each year for 6 years before a breach (*pre-breach year 6–pre-breach year 1*, backward), one for the year of a breach (*breach year 0*), and one dummy for each year for 6 years after a breach (*post-breach year 1–post-breach year 6*, forward). This analysis enables us to inspect the time trend of breach impacts before and after a breach. If unobserved factors in the trend of performance between the treatment and control groups drove differences, we would expect to observe such differences before a breach occurs. Table 4 reveals no significant effects in the pre-breach period, suggesting that

unobserved factors did not significantly affect hospital performance. Therefore, the parallel trend assumption is plausible. In the post-breach period, the negative effect of a breach is statistically significant for 3 years, but the effect diminishes in the subsequent years. The estimated coefficients of the dummy indicators are plotted in Supporting Information EC5.

4.3.2 Staggered DID Estimates. Recent advances in the econometrics literature suggest that using staggered treatment can result in biased inferences when treatment effects are heterogeneous (Callaway and Sant'Anna, 2021; Goodman-Bacon, 2021). In our research context, since breaches occur at different points in time, we conduct two tests to ensure that potential heterogeneity in breach impact over time has not introduced bias into the significance of the DID estimator.

First, we conduct Goodman-Bacon's weighted average analysis using a two-way fixed effects (TWFE) model that includes two indicators. One indicates whether a hospital is in the treatment group, and the other is a relative time indicator related to a breach time. In this model, the DID estimator from any TWFE model, where treatment occurs at different times, is a weighted average of all possible two-by-two DID estimators. Each DID estimator is obtained by comparing the treatment and control groups. Some comparisons use hospitals treated at a particular time as the treatment group and the untreated hospitals as the control group. Other comparisons use hospitals treated at two different times: (1) hospitals treated later as a control before their treatment begins, and (2) hospitals treated earlier as a control after their treatment begins. If treatment effects significantly vary over time, the DID estimator from this framework could be biased. The potential source of bias could come from the estimators where a group of early-treated hospitals acts as the control (Goodman-Bacon, 2021). Employing the *bacondecomp* library in Stata, the underlying

Table 4. Pre- and post-trends with the leads and lags models.

Variable	Admissions	Market share
Pre-breach year 6	−0.0319 (0.022)	−0.0012 (0.002)
Pre-breach year 5	−0.0100 (0.017)	−0.0002 (0.001)
Pre-breach year 4	0.0186 (0.015)	0.0032 (0.002)
Pre-breach year 3	0.0043 (0.014)	0.0013 (0.001)
Pre-breach year 2	−0.0039 (0.013)	0.0012 (0.001)
Pre-breach year 1	−0.0143 (0.013)	0.0006 (0.001)
Breach year 0	−0.0193 (0.011)	−0.0016 (0.001)
Post-breach year 1	−0.0291*** (0.012)	−0.0029** (0.001)
Post-breach year 2	−0.0322** (0.014)	−0.0028** (0.001)
Post-breach year 3	−0.0288** (0.014)	−0.0033** (0.001)
Post-breach year 4	−0.0096 (0.015)	−0.0030** (0.001)
Post-breach year 5	0.0098 (0.015)	−0.0020 (0.001)
Post-breach year 6	0.0080	−0.0012
Post-breach year 6	(0.014)	(0.001)
Controls	Yes	Yes
Year-fixed effect	Yes	Yes
Hospital-fixed effect	Yes	Yes
R ² (within)	0.1619	0.1280
R ² (overall)	0.5292	0.5327

Notes: $N = 15,031$. Number of hospitals = 1,766. Standard errors in parentheses are clustered at the hospital level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 5. Goodman-Bacon weighted average analysis.

Type of comparison	Weight	Average of DID estimator	
		Admissions	Market share
Earlier vs. later treated	0.058	−0.104	−0.009
Later vs. earlier treated	0.194	−0.013	−0.002
Later vs. always treated	0.137	−0.040	−0.003
Treated vs. untreated	0.611	−0.027	−0.004

DID estimators are derived. As shown in Table 5, the potential heterogeneity of breach impacts over time does not create significant bias for two reasons. First, the weights corresponding to comparisons where treated hospitals act as controls are lower (0.058, 0.194, and 0.137 in the first three rows) than the weight of the comparison where they act as treatments (0.611 in treated vs. untreated). Second, the average DID estimators are in the same direction and have no significant difference.

Next, to conduct a more in-depth validity test of our model, we analyze our data using the DID framework developed by Callaway and Sant'Anna (2021). While the Goodman-Bacon analysis can ensure the validity of our DID effects, the Callaway and Sant'Anna framework offers alternative DID estimates that help avoid any potential bias in interpreting the DID model. Callaway and Sant'Anna's DID framework primarily focuses on the disaggregated causal parameter known as the group-time average treatment effect in a specific time period, with a "group" being defined by the time when a hospital is treated. In Supporting Information EC6, "Breach time" represents the relative time to breach occurrence. Our analysis reveals no pretreatment effects on the outcomes of interest. The negative impacts of breaches on hospital performance become significant over the 3 years after a breach and then diminish. These negative trends in post-breach outcomes align with the patterns observed in the leads and lags analysis (Section 4.3.1), affirming the robustness of our main findings. The single- and group-aggregated effects are also significant, offering further support for our main findings. Figure 2 illustrates the pre- and post-breach trends from Callaway and Sant'Anna's DID framework.

4.3.3 Placebo Tests. To examine whether our results are due to spurious correlation by random chance, we conduct a robustness check on our estimates using randomization inference (Eftekhari et al., 2023; Imbens and Wooldridge, 2009). This involves the random shuffling of hospital temporal patterns based on equation (1), and the test is replicated 1,000 times. Figure 3 plots the distribution of the resulting coefficient values. Both distributions are centered on zero, and in comparison to the effects observed in our main specification, it is unlikely that our results are spurious. Table 6 shows the results of the randomization inference, the mean values of the randomization estimates, and their standard deviation, along with the main coefficients and their empirical p values from Tables 2 and 3. They provide evidence that our estimates are robust.

4.4 Disentangling the Mechanism of Breach Impacts

Our main analyses reveal that breached hospitals subsequently observe declines in both admissions and market share. We attribute these performance declines to operational disruption and the challenges arising from compliance with regulations. However, the impact of a breach on inpatient volume can be further dissected into two key aspects: operational inefficiency resulting directly from a breach or from subsequent remediation efforts, and reputational damage that may divert patients away from the affected hospitals.

To explore this mechanism in greater detail, we first include average inpatient days (Oh et al., 2018; Singh and Terwiesch, 2011) and ER visits (Gonçalves et al., 2023; Somanchi et al., 2022) as measures of operational efficiency. Next, to investigate the spillover effect possibly arising from reputational

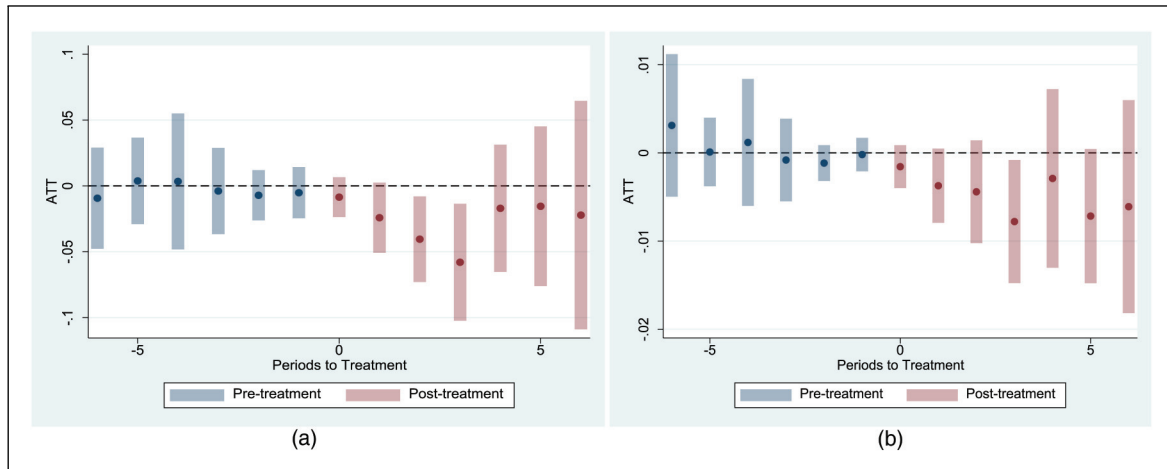


Figure 2. Pre- and post-trends with Callaway and Sant'Anna DID effects: pre- and post-trends with (a) admissions and (b) market share.

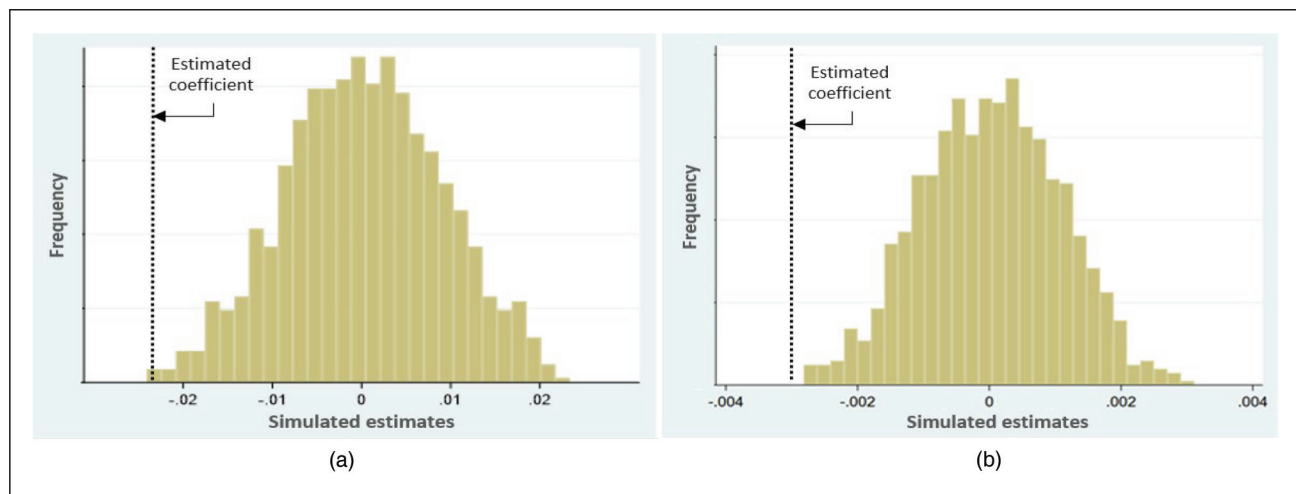


Figure 3. Placebo tests: breach impact coefficients in (a) admissions and (b) market share.

Table 6. Results from randomization inference with empirical p values.

	Admissions	Market share
Mean of random β	0.00002	0.00001
Standard deviation of random β	0.00905	0.00105
Replications	1,000	1,000
Estimated β	-0.0213	-0.0028
Empirical p value	0.0060	0.0288

damage, we incorporate the number of admissions in non-breached hospitals within the same HRR. It is important to note, however, that changes in admissions in non-breached hospitals may also result from hospital diversions/transfers from breached hospitals.

4.4.1 Operational Efficiency: Impact of Breaches on Inpatient Days and ER Visits.

Using our main DID-PSM models,

equations (1) and (2), we investigate the changes in a hospital's average inpatient days before and after a breach across the treatment and control groups. As shown in Table 7, breached hospitals experience a significant increase in average inpatient days (0.2537 at $p < 0.1$). This corresponds to an increase from an average of 5.5 days (see Table 1) to 5.75 days for inpatient stays. This increase is more pronounced for hospitals affiliated with a healthcare network (0.3557 at $p < 0.05$). Fully decentralized governance (score of 5) is associated with a 0.362 increase in average inpatient days. A 1% increase in the ratio of a hospital's cloud-based IT services over the total applications leads to an increase of 0.07 on average inpatient days (7.3987 at $p < 0.05$). Finally, a one-unit increase in the number of adopted IT security systems results in a decrease of 0.036 in average inpatient days (-0.0361 at $p < 0.1$).

Next, we examine the impact of a breach on a hospital's ER visits. As described in Table 8, we observe a significant decrease in a hospital's ER visits by roughly 8.8% (-0.0884

Table 7. Operational efficiency: impact of a data breach on average inpatient days.

Variable	(1)	(2)	(3)	(4)	(5)	(6)
Treat	0.2537** (0.129)	−0.2403 (0.185)	−0.2255 (0.152)	−0.2867 (0.180)	0.1996 (0.131)	0.0968 (0.226)
Treat × Network		0.3557** (0.188)				0.3231** (0.146)
Treat × Decentral			0.0724** (0.037)			0.0909** (0.045)
Treat × CloudITSvc				7.3987** (3.698)		8.1564*** (3.247)
Treat × ITSecuirty					−0.0361* (0.019)	−0.0609* (0.034)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Year-fixed effect	Yes	Yes	Yes	Yes	Yes	Yes
Hospital-fixed effect	Yes	Yes	Yes	Yes	Yes	Yes
R ² (within)	0.2088	0.2087	0.2101	0.2281	0.2093	0.2106
R ² (overall)	0.1118	0.1110	0.0962	0.0837	0.0952	0.0961

Notes: $N = 15,031$. Number of hospitals = 1,766. Standard errors in parentheses are clustered at the hospital level. A hospital's average inpatient days at time $t - 1$, is controlled. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 8. Operational efficiency: impact of a data breach on ER visits.

Variable	(1)	(2)	(3)	(4)	(5)	(6)
Treat	−0.0884*** (0.031)	−0.1000*** (0.030)	−0.0642** (0.029)	−0.0662** (0.028)	0.1439 (0.196)	−0.1064*** (0.031)
Treat × Network		0.1368 (0.109)				0.0506 (0.142)
Treat × Decentral			−0.0278 (0.032)			0.0368 (0.037)
Treat × CloudITSvc				−2.6962 (3.993)		−2.7834 (3.998)
Treat × ITSecuirty					−0.0360 (0.027)	−0.0174 (0.021)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Year-fixed effect	Yes	Yes	Yes	Yes	Yes	Yes
Hospital-fixed effect	Yes	Yes	Yes	Yes	Yes	Yes
R ² (within)	0.0394	0.0395	0.0395	0.0394	0.0391	0.0395
R ² (overall)	0.2838	0.2530	0.2517	0.2525	0.2529	0.2544

Notes: $N = 14,995$. Number of hospitals = 1,762. Standard errors in parentheses are clustered at the hospital level. A hospital's logged ER visits at time $t - 1$, is controlled. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

at $p < 0.01$). These results are consistent with prior work. In recent security research on ER visits, Neprash et al. (2024) found that ER volumes fell by 10–19% within the short term after an attack on rural and urban hospitals, respectively (Neprash et al., 2024). In another case study of a mid-sized medical center that experienced a ransomware attack, Gabbay-Benziv et al. (2023) observed a substantial decrease in ER visits, with a reduction of 5–10% over a short-term period of 11 weeks. Unlike these studies focusing on ransom attacks, our study examines changes in ER visits across a broader group of breaches over a long-term period. Given that the average logged number of ER visits is 6.929 (about

1,021 ER visits), an 8.8% reduction translates to an approximate annual decrease of 90 ER visits in the post-breach period (7.5 ER visits per month). However, none of the other interaction terms are significant. These findings suggest that ER visits are primarily associated with a breach, rather than being compounded by diverse organizational factors. This makes sense given the time-sensitive nature of ER visits, which may lead any breached hospital experiencing operational disruption, regardless of operational or technological structures, to divert the requests for ER visits. In addition, with the interaction with IT security systems, the effect becomes statistically insignificant, implying that security systems mitigate the adverse impact of a breach.

Table 9. Reputation effect: spillover of data breaches on admissions in non-breached hospitals.

Variable	(1)	(2)	(3)	(4)	(5)	(6)
Treat	0.0337*** (0.011)	0.0276*** (0.009)	0.0269*** (0.009)	0.0622*** (0.011)	0.0734*** (0.025)	−0.0093 (0.025)
Treat × Network		0.0506*** (0.013)				0.0358*** (0.014)
Treat × Decentral			0.0132*** (0.003)			0.0061** (0.003)
Treat × CloudITSvc				1.1738** (0.566)		2.1872*** (0.835)
Treat × ITSecurity					−0.0011 (0.003)	−0.0014 (0.003)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Year-fixed effect	Yes	Yes	Yes	Yes	Yes	Yes
Hospital-fixed effect	Yes	Yes	Yes	Yes	Yes	Yes
R ² (within)	0.8076	0.8077	0.8077	0.8077	0.8077	0.8081
R ² (overall)	0.9333	0.9332	0.9332	0.9332	0.9333	0.9332

Notes: N = 15,031. Number of hospitals = 1,766. Standard errors in parentheses are clustered at the hospital level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. The logged admissions of non-breached hospitals at time $t - 1$, is controlled.

4.4.2 Reputation Effect: Spillover Effect of Breaches to Non-Breached Hospitals. Given that a breach can cause both operational disruption and reputational damage, examining its spillover effect on nearby non-breached hospitals is meaningful. First, spillover may occur as patients choose non-breached hospitals due to concerns about the safety of their personal information and disrupted care quality resulting from a breach. Second, if a breached hospital is unable to deliver the required care, it may refer or divert patients to alternative facilities. Both situations raise the possibility of a spillover effect on non-breached hospitals within the same HRR. To investigate this effect, we incorporate the logged number of admissions from non-breached hospitals within the same HRR as an indicator of reputational damage.

Table 9 shows a notable trend: a breach at one hospital leads to a discernible increase in admissions at non-breached hospitals within the same HRR (0.0337 at $p < 0.01$). We observe that a breach in a hospital affiliated with a healthcare network or with decentralized governance exhibits positive moderating effects on the admissions of non-breached hospitals within the HRR (0.0506 and 0.0132 at $p < 0.01$, respectively). Additionally, when a hospital with cloud-based IT services is breached, non-breached hospitals experience an even greater increase in their admissions (1.1738 at $p < 0.01$). However, the interaction term with a breached hospital's IT security systems has no significant effect, suggesting that these security systems may positively impact patient retention through quick recovery.

Next, we explore other factors influencing patients' hospital choices, including market competition and health insurance plans. Since breaches pose secondary risks like identity theft and various frauds, patients are likely to seek for secure healthcare providers (Kox and Straathof, 2014). HIPAA breach notification rules⁹ have reduced information asymmetry in

security by alerting patients to breaches and subsequent potential frauds. However, given the wide regional variations in healthcare market competition, patients may face challenges in selecting hospitals based on secondary factors like security. For instance, in competitive markets with many hospitals, alternative providers can be easily found. But in less competitive markets, hospital choice is limited, making it difficult to find alternative providers that offer comparable healthcare services. Thus, we further analyze the impact of market competition by subdividing the treatment effects based on local market conditions.

First, we split the matched sample into two groups based on market competition, using an HHI value of 0.3¹⁰ in each HRR as the threshold. HRRs with an HHI greater than or equal to 0.3 are categorized as the high HHI group, representing less competitive markets, while those with an HHI less than 0.3 are included in the low HHI group, reflecting more competitive markets. As presented in Supporting Information EC7, breaches do not significantly impact hospital performance within the high HHI group, where a small number of hospitals compete. However, in the low HHI group, characterized by a large number of competing hospitals, breached hospitals experience a significant decrease in both admissions and market share (−0.0258 and −0.0026 at $p < 0.01$, respectively). In our dataset, competitive markets, on average, consist of 18 hospitals, whereas less competitive markets have an average of 5 hospitals. These statistics affirm that competitive markets provide patients with more alternatives.

Besides limited competition, hospital choices are further restricted by third-party payer systems where health services are not directly paid for by care recipients but rather by third-party payers (Guo et al., 2019). We disentangle the treatment effects across three types of health insurance plans:

HMO, PPO, and FFS plans. HMO plans offer significant economic incentives for patients to stay inside the HMO network, limiting choices based on personal preferences like security. FFS plans offer more flexibility in hospital choice but are generally more expensive. PPO plans are less restrictive than HMO plans but more restrictive than FFS plans. Supporting Information EC1 provides statistics summarizing hospital contracts with health insurance plans. As shown in Supporting Information EC8, HMO and PPO plans do not significantly moderate breach impact, but FFS plans have a negative moderating effect (-0.0319 at $p < 0.01$ and -0.0042 at $p < 0.05$). These results imply that when patients have the leeway to choose a hospital, they typically avoid hospitals that have experienced breaches (Adida and Bravo, 2019; Guo et al., 2019).

4.5 Additional Extension

4.5.1 External Attacks versus Internal Breaches. Breaches are typically categorized into two main types: internal and external, based on their source and motive (Kwon and Johnson, 2018). The matched sample consists of a total of 1,324 breaches, including 709 external (53.5%), 583 internal (44.0%), and 32 unknown (2.4%) incidents.¹¹ These distinct breach types can lead to varying impacts and require different remediation strategies. External breaches often trigger immediate disruptions and pose challenges for remediation due to the deliberate and criminal nature of the attacks. These malicious external breaches also introduce legal complexities, potentially resulting in secondary damages such as fraudulently obtained medical care, fake billing, and the emergence of black markets for stolen pharmaceuticals and equipment. Therefore, external breaches could have more pronounced negative effects on overall performance, such as admissions, compared to internal breaches. On the other hand, internal breaches, often resulting from insider mistakes or lapses in security management, appear to generate detrimental effects on operational efficiency, as reflected in average inpatient days and ER visits. Understanding how the impact of a breach varies with these breach types is fundamental for uncovering specific mechanisms.

To identify the effect of breach types, we incorporate a binary variable in equation (2) indicating whether a breach at time t is external (value of one) or not (value of zero). As shown in Table 10, the results consistently align with the main findings. We find that the coefficients of the interaction terms in columns (1) to (5) are larger than those from the prior analysis (without distinguishing the breach type). For instance, in Table 2, we report that admissions decrease by 2.1% (-0.0213 at $p < 0.01$) as compared to 2.7% (-0.0278 at $p < 0.01$) shown in Table 10 for external breaches. Similarly, the decrease in market share is enhanced from 0.28 to 0.43%. The magnitudes in ER visits and admissions in non-breached hospitals also slightly increase compared to those in Tables 8 and 9, respectively. The increased magnitudes indicate that external

breaches have a more significant effect than internal breaches. This makes sense, as external breaches typically receive more media coverage and public attention, leading to greater reputational damage and loss of trust among patients. However, the impact of external breaches on inpatient days remains limited, suggesting that inpatient care is more influenced by internal factors such as hospital capacity and patient management.

From a methodological standpoint, these analyses by breach types also demonstrate the robustness of our results. Considering external breaches as exogenous shocks and internal breaches as endogenous factors, the consistent impacts of external breaches emphasize that our results are not biased due to endogeneity issues.

4.5.2 IT Security Systems. Exploring the heterogeneous effects of breaches with IT security systems can provide valuable insights into IT security strategies. The interaction effects of IT security in Tables 2 and 3 demonstrate that IT security systems contribute not only to safeguarding against breaches but also to mitigating their adverse impacts. Given the varying functions within IT security systems, it is worthwhile to understand the distinct effects of these different functions for more nuanced IT security strategies. To explore this, we categorize them into four types: detection (e.g., intrusion detection and firewall), prevention (e.g., anti-virus and encryption), identity governance (e.g., user authorization/authentication), and recovery (e.g., backup). Summary statistics for IT security systems are provided in Supporting Information EC1.

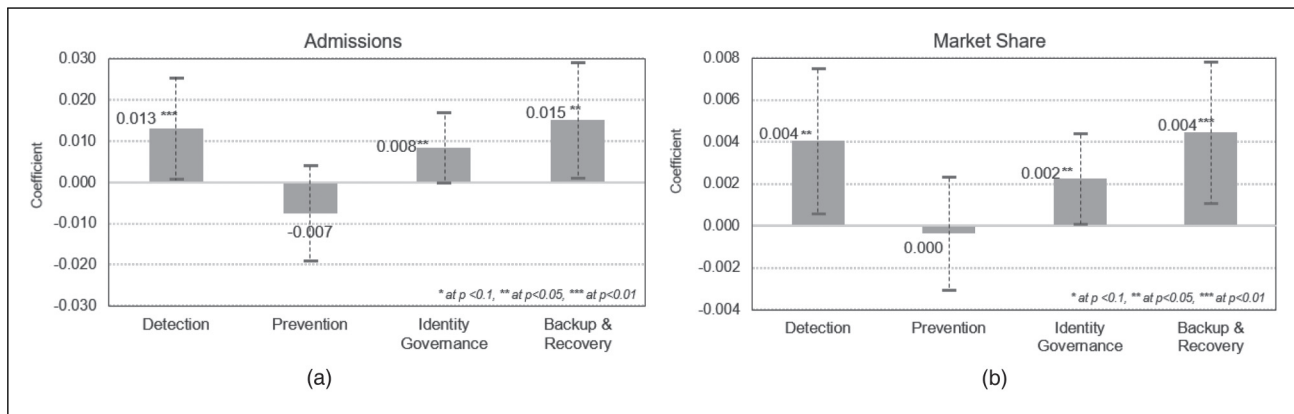
Figure 4 visually presents the results, with detailed findings available in Supporting Information EC9. Detection, identity governance, and backup & recovery systems significantly moderate the impact of data breaches, while prevention systems have no significant effect. These results are intuitive, as detection, identity governance, and backup & recovery systems enable quick restoration, reducing downtime and minimizing data compromise in the post-breach period. In contrast, prevention systems focused on averting breaches, have a limited post-breach impact on a hospital's performance. The implementation costs of detection, identity governance, and backup & recovery systems are relatively high, as they need to be well integrated into business processes, unlike prevention systems (e.g., antivirus and encryption), which typically require regular updates as off-the-shelf products.

4.5.3 Time Gaps Between Breach Occurrence and Public Announcement. We consider the possible time gap between the occurrence of a breach and its public announcement date. Operational disruption typically occurs immediately after a breach, as a hospital's systems are disrupted, followed by remediation efforts. In contrast, reputational damage occurs after a breach is publicly announced. Despite HIPAA and HITECH Act⁹ mandates requiring hospitals to promptly report breaches, announcements may be delayed for various reasons, such as slow breach detection or intentional procrastination to mitigate negative publicity. To assess the impact

Table 10. Impacts of an external breach on hospital performance.

Variable	Admissions (1)	Market share (2)	Inpatient days (3)	ER visits (4)	Admissions in non-breached hospitals (5)
Treat	−0.0137 (0.011)	−0.0034*** (0.001)	−0.3306** (0.162)	−0.1064 (0.143)	0.0065 (0.011)
Treat × External	−0.0278** (0.012)	−0.0043*** (0.001)	0.3774** (0.185)	−0.0894*** (0.019)	0.0382*** (0.016)
External	0.0038 (0.004)	0.0003 (0.000)	−0.0015 (0.310)	−0.0829 (0.087)	0.0773*** (0.016)
Controls	Yes	Yes	Yes	Yes	Yes
Year-fixed effect	Yes	Yes	Yes	Yes	Yes
Hospital-fixed effect	Yes	Yes	Yes	Yes	Yes
R ² (within)	0.1040	0.1179	0.2074	0.0395	0.8078
R ² (overall)	0.5170	0.4717	0.0779	0.2381	0.9315

Notes: Control variables are all used with year- and hospital-fixed effects. Standard errors in parentheses are clustered at the hospital level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

**Figure 4.** Moderating effects of IT security systems on hospital performance: the moderating effect of IT security on (a) admissions and (b) market share.

of the time gap between breach occurrence and announcement, we analyze three distinct pseudo-post-breach periods: 3, 6, and 9 months before the breach announcement date. This approach allows us to explore scenarios where a breach may have occurred months before its public disclosure. Detailed results are provided in Supporting Information EC10. We observe that the impact of a breach is marginally significant (-0.0143 at $p < 0.1$) only when the post-breach period starts 3 months before the announcement date, with no significant effect observed at 6 and 9 months prior. These findings affirm the adherence of most hospitals to HIPAA rules, reporting breaches within 3 months of discovery.

5 Implications and Conclusions

As hospitals' operating structures have expanded, organizational boundaries have become more permeable and blurred. Healthcare technologies have also continued to grow and

evolve, producing more data that are transferred to more devices in different locations, and thus creating new vulnerabilities. Given this shifting environment, we investigated the impact of data breaches on hospital performance and related moderating factors. Our study employed a quasi-experimental design using PSM and DID techniques to mitigate selection bias stemming from confounding factors. The PSM and DID models examined the treatment effect of a breach and the moderating roles of operational, technological, and market factors—such as network affiliation, governance structures, cloud-based IT services, IT security systems, market competition, and health insurance plans—on this treatment effect.

This study demonstrated that while a breach negatively impacts hospital performance, its impact is most pronounced in the 3 years after a breach and diminishes in the subsequent years. The negative effect of a breach can be explained from several aspects. First, operational challenges caused by a breach can lead to a reduction in hospital performance. We

analyzed this operational mechanism by incorporating additional dependent variables reflecting operational efficiency (i.e., average inpatient days and ER visits). The findings suggest an association between breaches and a significant reduction in ER visits, as well as an increase in inpatient days.

Second, the observed reduction in hospital performance also results from patient losses due to reputational damage. We gauged the market's response to a breach by investigating (1) the changes in a breached hospital's market share in the post-breach period, (2) the impact of a hospital's breach on the performance of non-breached hospitals within the same HRR, (3) the moderating effect of market competition, and (4) how the impact of a breach varies with health insurance plans that influence patients' hospital choices. We find that a hospital's breach decreases its market share within an HRR but boosts the performance of non-breached nearby hospitals. Furthermore, a breach has a limited effect on highly concentrated markets where only a few hospitals operate, whereas it has a significantly greater impact on less concentrated markets with more alternatives. Considering health insurance plans with third-party payers, hospitals offering FFS plans, which allow flexible hospital choice, experience a larger negative impact than hospitals offering HMO or PPO plans that restrict hospital choice. These results indicate that in competitive markets with comparable alternatives or flexible healthcare plans, hospitals are subject to market forces reflecting patient preferences for information security.

We further identified the moderating effects of operational and technological factors. From an organizational standpoint, a hospital's network affiliation intensifies the negative impact of a breach on hospital performance. Member hospitals within a healthcare network strive to harness intra-organizational collaboration for value creation and competitive advantage. However, this collaboration heightens interdependence, rendering members more susceptible to cyberattacks. In this expanded operating structure, governance plays an important role in breach resilience. Our results reveal a negative interaction between decentralized governance and breach impact on hospital performance. This implies that decentralized governance hinders a cohesive and rapid response to breaches, thereby increasing overall security risks due to divergent prioritization of mitigation and compliance among hospitals.

From a technological perspective, our findings indicate that hospitals heavily reliant on cloud-based IT services experience more pronounced negative impacts. The extended IT service structure increases the attackable surfaces, complicating the identification of breach responsibility and swift action. However, breach impact can be mitigated when hospitals are well-equipped with IT security systems. Detailed analyses of the four types of IT security systems reveal that detection, identity governance, and recovery systems consistently mitigate the negative impact of breaches. Prevention systems show only partial effectiveness in mitigating the impact of breaches, as their primary role is to deter breaches rather than mitigate their consequences. Nevertheless, our results suggest that

overall IT security systems across all four types can contribute significantly to breach resilience.

Our findings make important theoretical and managerial contributions to the OM and security literature. First, while previous research has predominantly focused on the antecedents of breaches for ex-ante prevention strategies, there is a notable gap in studies addressing the consequences, mitigation, and resilience strategies following breaches. This study addresses this gap by identifying the mechanisms through which breaches lead to decreased performance, considering two perspectives. From an operational standpoint, we find that breaches cause operational disruptions, negatively impacting efficiency. From a market perspective, we show that a hospital breach has a spillover effect, increasing inpatient volume in neighboring non-breached hospitals. Our study also provides evidence that hospitals shoulder a breach with little loss when operating in highly concentrated markets with few competitors and when patients' insurance plans do not allow flexible hospital choice. Second, our findings delineate the institutional conditions under which operating and governance structures exacerbate breach impacts. Healthcare networks have become a prevalent institutional form in the USA, offering integrated healthcare services through collaborations with various specialized care providers (McConnell, 2020). Our study underscores the critical implications of this evolving care delivery model for understanding the relationship between operating structure and security risk management. We observe that interdependent and interconnected operating structures create more exposed vulnerabilities to breaches, resulting in notable performance declines. Moreover, decentralized governance, which grants member hospitals autonomy in decision-making regarding services and IT systems, appears to negatively impact performance following a breach. Third, although prior security research has employed stylized analytical models to assess the economic outcomes of IT security investment or sourcing strategies (Bensoussan et al., 2020; Cezar et al., 2017), our study adopts an empirical approach to quantify how cloud-based IT services and specific IT security systems mitigate the impact of a breach. Our results indicate that hospitals utilizing cloud-based IT services experience greater performance declines, while those equipped with robust IT security systems experience smaller decreases.

From a policy and managerial perspective, our findings provide operational and economic reasoning for hospital managers and policymakers to make relevant security decisions that enhance breach resilience. Expanded and autonomous operating structures with network affiliation and decentralized governance exacerbate the impact of breaches. Although healthcare networks with decentralized governance may enhance healthcare services with agility, they impede breach handling, creating operational and legal challenges. These findings suggest that hospital managers must carefully weigh the trade-offs between effective care coordination and risk management to ensure resilience against cyberattacks.

Our analyses of healthcare market mechanisms imply that consumerization via flexible hospital choice can create opportunities for utilizing market-based instruments for information security. Thus, security regulations need to be more flexible and tailored, taking into account competitive dynamics and institutional referral patterns rather than imposing across-the-board regulations. Policymakers should carefully examine markets with limited competition and adjust security strategies to align with the characteristics of hospitals and healthcare markets. For example, in less competitive markets where a few hospitals dominate, regulatory intervention may need to take a stick approach (e.g., penalties) because such dominant hospitals dull market forces reflecting security preference, allowing them to endure a breach with little loss. On the other hand, a carrot approach (e.g., incentives) could encourage small hospitals with limited market power and fewer resources to make security investments. Given that smaller hospitals may lack the human capital or resources for significant security investments, incentivizing voluntary security actions through the carrot approach could attract patients seeking enhanced security, leading to increased market share and brand-building. Policymakers should consider incentives to promote information sharing related to security technologies, privacy policies, and breach incidents among healthcare organizations and with federal agencies (Yaraghi, 2016).

This study has some limitations. First, our analyses are conducted at the hospital level, which does not provide implications for patient-level characteristics, such as medical conditions, which would be valuable for hospital choice. Second, given the unique characteristics of the US healthcare sector (e.g., government funding and federal breach notification rules), our findings may not be readily generalized to other industries or countries facing different levels of security regulation and market characteristics (e.g., market trajectories and competition). Lastly, the spillover effect of breaches could result from changes in hospital choices due to both reputational damage and hospital diversions/transfers. Given the observational nature of our data, our analysis is limited in disentangling these two effects.

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ORCID iD

Juhee Kwon  <https://orcid.org/0000-0002-3097-0185>

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Notes

1. In Table 3, the dependent variable is the log-transformed number of admissions. The treatment coefficient of -0.0213 corresponds to an approximate 2.13% decrease (calculated as $1 - e^{-0.0213} \approx 0.021$) in admissions. With an average of 2,858 admissions in the total sample, this translates to an average loss of 60 admissions per hospital ($2,858 \times 0.0213$).
2. Given that a hospital's revenue per admission is \$27,023, the average revenue loss amounts to approximately \$1.6m, calculated by multiplying \$27,023 by 60.
3. As required by Section 13402(e)(4) of the HITECH Act, the HHS has posted a list of breaches affecting 500 or more individuals. See https://ocrportal.hhs.gov/ocr/breach/breach_report.jsf.
4. HRRs represent regional healthcare markets with a minimum overall population of 120,000. An HRR includes an area for tertiary medical care that is referred to primary medical caregivers.
5. Decentralized governance is measured using Bazzoli et al.'s (1999) taxonomy, a standard in the AHA database. The scale ranges from (1) high centralization in services, physicians, and insurance to (5) full decentralization with autonomous member hospitals (AHA, 2017).
6. Based on CMS guidelines, this study includes a risk-adjusted readmission rate, which calculates the proportion of unplanned readmissions to an acute-care hospital within 30 days post-hospitalization.
7. CMS provides information about healthcare services provided through arrangement or referral. This captures a hospital's healthcare service interconnection with other healthcare providers, which reflects the complexity of healthcare services.
8. Several matching distances were tested. The maximum matching distance (0.10) was selected from within the range over which the t -tests showed no difference between the treatment and control groups.
9. HIPAA breach notification rules force a breached hospital to notify four entities (i.e., patients, business associates, HHS, and local media) within 60 days.
10. We found the 0.3 criteria by conducting multiple analyses with a range from 0.1 to 0.5 HHI.
11. The total sample includes 609 internal (44.4%), 730 external (53.2%), and 32 with unknown sources (2.3%) out of a total of 1,371 breaches.

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